

Collins ep reverse-unitarity supporting numerical evidence: v0.5.1

Observed execution and evidence categories

CHECKS_PASS_WITH_EXPLICIT_SCIENTIFIC_QUALIFICATIONS. Both independent frozen-source official r00-r07 workflows completed STAGES_PASS. The official seed verifiers for 1729 and 92741 each report CHECKS_PASS with 182 checks and zero failures; the official two-seed comparator reports CHECKS_PASS. This report separates those automated acceptance records from new numerical checks, exact readback, retained evidence and the unresolved literature comparisons. The numerical records do not erase the analytical K11 bulk discrepancy or the declared physical scope limits.

Software selftests, native integration, exact supporting checks, numerical amplitude comparisons, historical accepted evidence and final official acceptance are separate records. Their counts cannot be added to obtain a literature equation count. The companion analytical report preserves the 85 historical equation groups.

Record	Observed result	Scope
Validator selftest	111 tests; 0 failures; 0 errors	Software tests only
Reusable native regression	REUSABLE_NATIVE_SETUP	286 unchanged evidence entries; release, runtime and adoption identities match
Native exact regression checks	10 exact zero differences	Cut volume, eikonal moment and actual Kira identities
Complete Kira-map probes	Seeds 1729, 92741	Mutation includes implicit master identities; preserved regression evidence
Period-disabled negative control	Exit 1; no local PASS receipt	Exact export rejection is an expected negative control

Setup: [record](#); selftest command and log: [record](#).

Workflow / report	Requested scope	Status	Observed detail
20260914T173424Z-c452c5a14479	r03	DEVELOPMENT_PASS	
20260914T173549Z-92c3deda2cea	r06	DEVELOPMENT_PASS	
20260914T181858Z-de31606fa32d	r07	STAGES_PASS	
20260914T182256Z-e1c5081337e7	r07	STAGES_PASS	
ru-1729.json	Official seed or pair gate	CHECKS_PASS	{'PASS': 182}
ru-92741.json	Official seed or pair gate	CHECKS_PASS	{'PASS': 182}
ru-pair.json	Official seed or pair gate	CHECKS_PASS	Both seeded report identities and retained replay accepted

The reused setup establishes that the repaired native boundary and complete reduction-map mutation work for the tested pilot family. It is not a replacement for the two complete fresh scientific workflows and their official seed and dependency checks.

Official seed comparisons and native dependency probes

Seed	Official check rows	Accepted-engine scalar targets	Native probe gates	Status
1729	182	167	3	CHECKS_PASS
92741	182	167	3	CHECKS_PASS

Seed	Perturbation	Exact multiplier	Changed exports	Status
1729	kira_real_rules	11/6	25	PASS
1729	real_master	11/6	25	PASS
1729	virtual_master	11/6	5	PASS
92741	kira_real_rules	4/3	25	PASS
92741	real_master	4/3	25	PASS
92741	virtual_master	4/3	5	PASS

These are the actual protected official probes, distinct from the earlier scoped virtual diagnostic. Each mutates a copied native reduction/master result, keeps contraction recipes fixed, and executes the required downstream stages. The complete audited Kira-map mutation includes implicit master identities. A change count reports responding scalar exports; it is not a count of independent physical diagrams or literature equations.

The 167 accepted-engine scalar comparisons and the remaining native/replay/coverage/probe gates belong to the 182 official check rows per seed. The 85 analytical groups and their preserved discrepancies use a different grouping. Official CHECKS_PASS is an automated workflow/evidence result; the declared physics qualifications remain in force.

[1729 kira_real_rules](#); [1729 real_master](#); [1729 virtual_master](#); [92741 kira_real_rules](#); [92741 real_master](#); [92741 virtual_master](#);

Actual r00 outputs and retained independent helicity evidence

The current report binds newly generated r00 outputs to a previously executed direct comparison by exact file identity. It does not rerun that numerical evaluator or generate new MadGraph amplitudes. The prior comparison evaluated 36 physical points across two seeds, comprising 36 UU and 24 quark transverse-spin scalar targets per comparison route. Four actual native Born/real amplitude and tensor/function files and the complete retained comparison manifest are checked by the binding script.

Binding result: RETAINED DIRECT BINDING APPLIES TO IDENTICAL OUTPUTS. The receipt covers 2 actual workflows and records each run identity and source/output hash.

Comparison route	Targets	Maximum absolute residual	Mismatches
fresh native function vs fresh unsquared	60	2.220446049250e-16	0
fresh native function vs retained MG	60	2.775557561563e-16	0
fresh unsquared vs retained MG	60	1.942890293094e-16	0

Status is NUMERICAL AGREEMENT. Each tolerance is $2e-10 + 2e-7 \max(\text{abs}(\text{candidate}), \text{abs}(\text{reference}))$. Floating-point agreement is not symbolic equality. The original candidate/reference values, tolerances and pointwise residuals remain linked in the preserved comparison record.

The UT target uses complex helicity interference and incoming/outgoing off-diagonal spin-density matrices. Native HELAS and candidate spinor phases are separately calibrated; completeness, spin reversal, zero-spin and compensated phase controls are retained. The SIDIS conversion uses $q=l\text{-lprime}$ and $k1=p\text{-prime}$. These photon-parton invariants are distinct from electron-parton Mandelstam variables. Squared amplitudes include the charge, electromagnetic and strong couplings and initial spin/color averages, but no flux or phase-space factors. Hadron mass, z and Fourier/Collins-moment conventions are outside this benchmark.

Retained control	Maximum absolute residual
born tensor vs scalar UT	6.638929877565e-18
imaginary residual max	0.000000000000e+00
new helicity phase or reversal	2.227212004505e-16
new spinor completeness	2.131628207280e-14
ward gluon	3.477336605596e-29
ward photon	3.647173215880e-29

New-output identity receipt: [record](#); executed comparison: [record](#); pointwise values: [record](#). Identity propagation applies only to the four matched four-dimensional tree artifacts. It does not test epsilon-dependent operator numerators, HF gauge completion, virtual integration, measured distributions, rapidity subtractions or the final hadronic observable.

HF current and cut comparison

A fresh FeynCalc trace is compared with a separate explicit Clifford/hat algebra and with the conjugate cut. Three rational physical points, three channels and the ϵ^0 and ϵ^1 coefficients yield 18 rows. The evaluator uses 40 decimal digits and tolerance $1e-30$; exact rational-point residuals are retained separately. These are operator/current checks, without a new MadGraph claim or a claim of complete phase-space integration.

Rational point	Channel	epsilon order	Absolute residuals	Exact point residuals	Classification
{2/5, 1/3, 1}	a	0	0, 0	0, 0	NUMERICAL AGREEMENT
{2/5, 1/3, 1}	a	1	0, 0	0, 0	NUMERICAL AGREEMENT
{2/5, 1/3, 1}	b	0	0, 0	0, 0	NUMERICAL AGREEMENT
{2/5, 1/3, 1}	b	1	0, 0	0, 0	NUMERICAL AGREEMENT
{2/5, 1/3, 1}	c	0	0, 0	0, 0	NUMERICAL AGREEMENT
{2/5, 1/3, 1}	c	1	0, 0	0, 0	NUMERICAL AGREEMENT
{3/7, 4/9, 9/4}	a	0	0, 0	0, 0	NUMERICAL AGREEMENT
{3/7, 4/9, 9/4}	a	1	0, 0	0, 0	NUMERICAL AGREEMENT
{3/7, 4/9, 9/4}	b	0	0, 0	0, 0	NUMERICAL AGREEMENT
{3/7, 4/9, 9/4}	b	1	0, 0	0, 0	NUMERICAL AGREEMENT
{3/7, 4/9, 9/4}	c	0	0, 0	0, 0	NUMERICAL AGREEMENT
{3/7, 4/9, 9/4}	c	1	0, 0	0, 0	NUMERICAL AGREEMENT
{5/8, 7/11, 4}	a	0	0, 0	0, 0	NUMERICAL AGREEMENT
{5/8, 7/11, 4}	a	1	0, 0	0, 0	NUMERICAL AGREEMENT
{5/8, 7/11, 4}	b	0	0, 0	0, 0	NUMERICAL AGREEMENT
{5/8, 7/11, 4}	b	1	0, 0	0, 0	NUMERICAL AGREEMENT
{5/8, 7/11, 4}	c	0	0, 0	0, 0	NUMERICAL AGREEMENT
{5/8, 7/11, 4}	c	1	0, 0	0, 0	NUMERICAL AGREEMENT

Final integrated r00 binding: PASS

The final HF angular diagnostic uses an actual full-D momentum/hat trace and performs the rank-one Fourier phase and Beta-angle transform before applying the declared common scalar/rank-one operator continuation. Its difference from the nominal physical-plane density is $-\epsilon^2 (1-u)^2 (CA-2CF+2CF \ v)/(1-v)^2$. The ϵ^0 , ϵ^1 and one-loop finite radial shifts vanish exactly on the uniform physical endpoint class. This is not all-epsilon equality or a generic contact theorem. The invalid earlier postprocessing diagnostic is preserved and superseded explicitly by angular-003.

Pointwise current/cut data: [record](#) ; final source/output binding: [record](#). A separate fresh complex-current check has 12 finite-damping rows, including 8 with nonzero imaginary parts. It independently reverses and Dirac-conjugates the right current and conjugates propagator/axial prescriptions. The complex coefficient $2+3i$ tests the Hermitian interference sign. All recorded exact residuals vanish; the numerical display uses 40 digits and tolerance $1e-30$. This is direct-current evidence, not MadGraph, and finite damping does not prove generic endpoint convergence or a generic physical complex-F identification.

Complex values, conjugated right current and interference: [record](#).

Independent native pilot, virtual and response readback

The following entries are exact algebra and native execution evidence, not floating-point acceptance counts. Each check reads actual exports from the indicated protected official development run. Native derivations were executed by the workflow; the supporting checks then independently evaluate the measured beta integral, Gamma expansion, reduction and distribution action.

Supporting check	Observed result	Scope / settings
Inclusive quark pilot	14 exact zero differences; 5 coefficient arithmetic/depth checks	Actual Kira map; normalized cut master through ϵ^2 ; angular beta moment; delta, D0, D1, regular terms; four polynomial test convolutions
Ordinary virtual current	9 exact zero differences; 8 coefficient arithmetic/depth checks	$D=d=4-2\epsilon$; actual Kira current coefficient; normalized bubble through ϵ ; separate UV residues; IR subtraction and finite hard
Scoped virtual master probe	8 exact zero response residuals	Protected native mutator scales a copied actual B by 11/6; same copied source recipes; protected stage execution; probe poles retained
Own preliminary virtual checker	One failed check retained, then corrected	Kira d conversion was applied but numerator D conversion omitted. This was a review-checker defect, not a candidate failure.

The pilot is an inclusive quark longitudinal moment. It does not by itself establish fixed-z measured beam, fragmentation, HF or cone factors. The scoped virtual probe reruns r06 on copied prefix outputs and is not a second full official workflow or a complete official dependency campaign.

Actual pilot readback: [record](#); actual virtual readback and probe: [record](#); preserved checker failure: [record](#).

Additional current record	Status
HF fresh generation, angular proof and 18 numerical rows	FRESH_GENERATING_ANGULAR_AND_18_NUMERICAL_CUT_ROWS_PASS
Final native soft review	FINAL_NATIVE_SOFT_REVIEW_PASS
Actual Kira raised measurement transport	EXACT_NATIVE_KIRA_RAISED_MEASUREMENT_TRANSPORT_CHECKS_PASS
HF complex-current evidence receipt	PASS

Measured distributions and independent convolutions

The focused readback independently checks the actual native coefficient recipes, measured Fourier/sphere normalization and distributional action. It reads actual exports and no final comparison target. Polynomial tests are 1, z and z^2 ; lower limits are 0, $1/4$ and $2/5$. Eta is expanded before epsilon. Certificate arithmetic and independent test-function checks are listed separately.

Check family	Observed counts	Arithmetic / settings
Native coefficient recipes	{'EXACT': 282, 'MISMATCH': 0}	Exact symbolic contraction arithmetic; not a count of independent literature equations
Distribution / measure consequences	{'EXACT': 27, 'NUMERICAL_AGREEMENT': 18, 'MISMATCH': 0}	27 exact focused identities and 18 separate numerical finite-fragment convolutions
Finite-fragment convolutions	18 NUMERICAL AGREEMENT; no mismatches	90 decimal digits; absolute tolerance $1e-65$; $CF=4/3$; lower limits $1/4$ or $2/5$; tests $1, z, z^2$

Finite-fragment test	Numerical residual	Tolerance
dqq finite convolution z^0 , lower= $1/4$	0.0	absolute $1e-65$
dqq finite convolution z^1 , lower= $1/4$	0.0	absolute $1e-65$
dqq finite convolution z^2 , lower= $1/4$	0.0	absolute $1e-65$
dqq finite convolution z^0 , lower= $2/5$	0.0	absolute $1e-65$
dqq finite convolution z^1 , lower= $2/5$	0.0	absolute $1e-65$
dqq finite convolution z^2 , lower= $2/5$	0.0	absolute $1e-65$
dgq finite convolution z^0 , lower= $1/4$	0.0	absolute $1e-65$
dgq finite convolution z^1 , lower= $1/4$	0.0	absolute $1e-65$
dgq finite convolution z^2 , lower= $1/4$	0.0	absolute $1e-65$
dgq finite convolution z^0 , lower= $2/5$	0.0	absolute $1e-65$
dgq finite convolution z^1 , lower= $2/5$	0.0	absolute $1e-65$
dgq finite convolution z^2 , lower= $2/5$	0.0	absolute $1e-65$
collins finite convolution z^0 , lower= $1/4$	0.0	absolute $1e-65$
collins finite convolution z^1 , lower= $1/4$	0.0	absolute $1e-65$
collins finite convolution z^2 , lower= $1/4$	0.0	absolute $1e-65$
collins finite convolution z^0 , lower= $2/5$	0.0	absolute $1e-65$
collins finite convolution z^1 , lower= $2/5$	0.0	absolute $1e-65$
collins finite convolution z^2 , lower= $2/5$	0.0	absolute $1e-65$

Numerical zero after finite-precision quadrature is NUMERICAL AGREEMENT. Exact residuals are stored separately for shifted lower-limit plus actions, local vector/tensor moments, Fourier/Gamma normalization and the actual-master transversity angular finite shift. The one-loop spin continuation qualification does not imply an all-order tensor identity.

Original checked native scope: These checks bind the actual final frozen-source official run A native r00-r07 outputs. The final two-seed comparison and dependency probes remain separately reported obligations.

Final-output binding: The focused checker directly read the actual final official run exports and masters; their hashes are recorded in its result.

Focused exact and numerical data: [record](#).

Independent soft angular quadrature and normalization

A review-only derivation restores the longitudinal $da/a=dy$ measure from the physical two-cut density and checks the global beam-jet eikonal current against the common measured scalar sphere. Fourteen exact geometry and normalization identities are checked. An independent radial quadrature at transverse dimension 3 ($\epsilon=-1/2$) compares the conditional angular integral with its Gamma expression. It does not use candidate finite coefficients as inputs.

epsilon	eta	Direct quadrature	Absolute residual	Tolerance
-0.5	0.5	3.141592653589793238462643383279	1.24440052651e-35	1.0e-30
-0.5	1.0	2.467401100272339654708622749969	3.11150763893e-61	1.0e-30
-0.5	1.5	3.141592653589793238462643383279	1.24440052651e-35	1.0e-30

All three numerical comparisons agree at tolerance $1e-30$, using 60 decimal digits and the radial substitution $r=x^4$, split at 0, $1/4$, $3/4$ and 1. The largest absolute residual is $1.24440052651e-35$. These convergent dimension/regulator settings test the conditional angular measure; continuation and eta-before-epsilon cancellation are separately exact obligations.

The initial source review identified the need to display the actual eikonal current, conditional angular Jacobian, longitudinal restoration, regulator conventions and pre-epsilon eta gate. The initial snapshot is preserved. The separate final-source review binds those corrections to actual official r00/r03 output and checks five exact soft contraction/eta/UV identities. A standalone angular quadrature is not a substitute for that native binding.

Executable quadrature and exact checks: [record](#); source snapshot review: [record](#).

Retained campaign, runtime identities and reproduction

Historical seed	Derivation checks	NLO checks	Historical status
1729	{'PASS': 1337}	{'PASS': 384}	CHECKS_PASS
92741	{'PASS': 1337}	{'PASS': 384}	CHECKS_PASS

The historical v0.4.0 pair status is CHECKS_PASS. These are retained campaign records, not newly generated v0.5.1 amplitudes. Each successful official seed and the official pair comparator also freshly replay the protected retained evidence comparator and bind its output hash. That replay regenerates no MadGraph/amplitude sources. Shared algorithms and identical outputs are disclosed and do not establish independent operator derivations.

Native dependency	Exact release / executable / source identity
Frozen validator release	7bd929a50ea81b8af7d58c91063205d5047cc1bf60b7aa6c449d7661d2f79f0f
fermat	cca7f136820e21be0b9f8a23a0efe9289ce62e59d0e88702150809d9815dfc2d
feynarts_root	3c23f3f39ae06841cba8373277265882d48a1888392f712948a3697661a357c4
feyncalc_root	db35d60226cca51372c4a34e4e3fde4a992323828d20c8e7761db0fe41066e9a
kira	d57a64ffe5afd3d1174f8805da78b5134f9636359c821d9523ca1b757bf83fbf
polymake	5ee2e7de26194b575acb45e711905149fd7206e1184ae86a11f0eb5b31fcc690
python	a2f33a6e006989270f4340528eb61f8f97366e00a5d1b602ac8672ea44fc56ae
subtropica_root	f25e7ea6cdfc8557f42c348d174ab49ad20714e35000609fc8316d9b8bca47e4
wolfram_kernel	4f3f249b461dc8cf1b259eb86849cb3ff36ca7b54ac9266d5c7b2cab01bcfab4

Leading-power Born plus one loop, with the inherited narrow-cone and physical HF endpoint qualifications. Generic contact coefficients, R=1, unrestricted nonsingular fixed-order NLO, closed full twist-three evolution and NNLO are not certified. Fitted functions remain symbolic and Figure6 remains paused. Shared generating source and retained evidence are disclosed; no analytical equality alone certifies the final physical prediction.

Reproduce this supporting report with the command in the Markdown file. It reads preserved results, verifies hashes and writes to a fresh output directory. Check-generating scripts and exact native execution commands are linked in each receipt. The report builder never supplies formulas to the producing engine. Full report input hashes are recorded in the companion JSON.